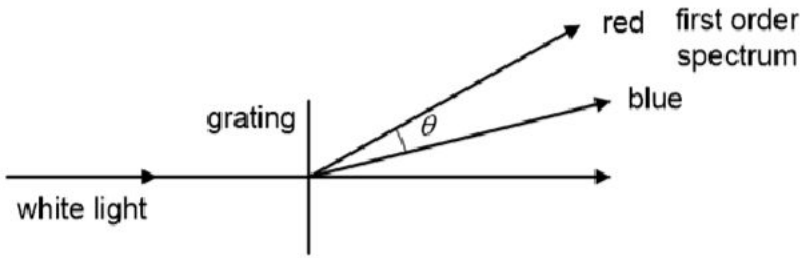
13 A compact spectrometer uses a diffraction grating containing 4.0×10^5 lines per metre. White light with wavelengths from 420 nm to 680 nm is incident normally on the grating.



What is the angular separation between the blue and red ends of the first-order spectrum?

A 6.1º **B** 9.7º **C** 15.8º **D** 25.5º

Ans: A

For blue light:

$d \sin\theta_B = n\lambda_B$

$$\sin\theta_B = \frac{420 \times 10^{-9}}{\frac{1}{4.0 \times 10^5}}$$

$\theta_B = 9.67^\circ$

For red light:

$d \sin\theta_R = n\lambda_R$

$$\sin\theta_R = \frac{680 \times 10^{-9}}{\frac{1}{4.0 \times 10^5}}$$

$\theta_R = 15.8^\circ$

Separation, $\Delta\theta = 15.8 - 9.67 = 6.13^\circ$